1D Burgers equation

$$\partial_t u = -\frac{1}{2}\partial_x u^2 + \nu \partial_{xx} u + f$$

We want to solve this in Fourier space:

$$\partial_t \hat{u}_k = -\frac{ik}{2}\widehat{(u^2)}_k - \nu k^2 \hat{u}_k + \hat{f}_k$$

We want to solve the dissipation explicitely $\Longrightarrow$ Ansatz:

$$\hat{u}_k(t) = \hat{\tilde{u}}_k(t) e^{-\nu k^2 t}$$

$\Longrightarrow$

$$\partial_t \hat{\tilde{u}}_k = \left(-\frac{ik}{2}\widehat{(u^2)}_k + \hat{f}_k \right) e^{\nu k^2 t}$$

Now we have no dissipation anymore at the expense of a time-dependent right hand side.

Euler Step

Euler step in general:

$$u^{(1)} = u^{(0)} + \Delta t L[u^{(0)}, t = t^{(0)}]$$

Now for our problem:

$$\hat{\tilde{u}}_k^{(1)} = \hat{\tilde{u}}_k^{(0)} + \Delta t \left(-\frac{ik}{2} \widehat{\left(u^{(0)^2} \right)}_k + \hat{f}_k^{(0)} \right) e^{\nu k^2 0}$$

or in our original $\hat{u}_k(t)$:

$$\hat{u}_k^{(1)} = e^{-\nu k^2 \Delta t} \left[\hat{u}_k^{(0)} + \Delta t \left(-\frac{ik}{2} \widehat{\left(u^{(0)^2} \right)}_k + \hat{f}_k^{(0)} \right) \right]$$

Pseudospectral steps

- Initialisation
  - initial condition in real space $u_0(x)$
  - FFT $\Longrightarrow \hat{u}_k^{(0)}$
  - same for forcing $\Longrightarrow \hat{f}_k^{(0)}$
- Euler steps
  - aliasing $\hat{u}_k^{(0)}$
  - transform $\hat{u}_k^{(0)}$ in real space $\Longrightarrow u^{(0)}(x)$
  - square it and store it in temporary field: $\mathrm{tmp}(x) = \frac{1}{2} {u^{(0)}}^2(x)$
  - FFT of $\mathrm{tmp}(x) \Longrightarrow \widehat{\mathrm{tmp}}_k$
  - $\hat{u}_k^{(1)} = e^{-\nu k^2 \Delta t} \left[\hat{u}_k^{(0)} + \Delta t \left(-\frac{ik}{2} \widehat{\mathrm{tmp}}_k + \hat{f}_k^{(0)} \right) \right]$

stable RK2: Heuns method

Heuns method in general:

$$\begin{aligned} u^{(1)} &= u^{(0)} + \Delta t L[u^{(0)}, t = t^{(0)}] \\ u^{(2)} &= \frac{1}{2}u^{(0)} + \frac{1}{2}u^{(1)} + \frac{1}{2}\Delta t L[u^{(1)}, t = t^{(1)}] \end{aligned}$$

Now for our problem:

$$\begin{aligned} \hat{u}_k^{(1)} &= e^{-\nu k^2 \Delta t}\left[\hat{u}_k^{(0)} + \Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(0)^2}\right)}_k + \hat{f}_k^{(0)}\right)\right] \\ \hat{\hat{u}}_k^{(2)} &= \frac{1}{2}\hat{u}_k^{(0)} + \frac{1}{2}\hat{\hat{u}}_k^{(1)} + \frac{1}{2}e^{\nu k^2 \Delta t}\Delta t\left[-\frac{ik}{2}\widehat{\left(u^{(1)^2}\right)}_k + \hat{f}_k^{(1)}\right] \end{aligned}$$

or

$$\hat{u}_k^{(2)} = \frac{1}{2}e^{-\nu k^2 \Delta t}\hat{u}_k^{(0)} + \frac{1}{2}\left[\hat{u}_k^{(1)} + \Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(1)^2}\right)}_k + \hat{f}_k^{(1)}\right)\right]$$

Pseudospectral steps

- aliasing $\hat{u}_k^{(0)}$
- transform $\hat{u}_k^{(0)}$ in real space $\Longrightarrow u^{(0)}(x)$
- square it and store it in temporary field: $\mathrm{tmp}_1(x) = \frac{1}{2} {u^{(0)}}^2(x)$
- FFT of $\mathrm{tmp}(x) \Longrightarrow \widehat{\mathrm{tmp}}_k$
- $\hat{u}_k^{(1)} = e^{-\nu k^2 \Delta t} \left[\hat{u}_k^{(0)} + \Delta t \left(-\frac{ik}{2} \widehat{\mathrm{tmp}_1}_k + \hat{f}_k^{(0)} \right) \right]$
- aliasing $\hat{u}_k^{(1)}$
- transform $\hat{u}_k^{(1)}$ in real space $\Longrightarrow u^{(1)}(x)$
- square it and store it in temporary field: $\mathrm{tmp}_2(x) = \frac{1}{2} {\tilde{u}^{(1)}}^2(x)$
- FFT of $\mathrm{tmp}_2(x) \Longrightarrow \widehat{\mathrm{tmp}_2}_k$
- $\hat{u}_k^{(2)} = \frac{1}{2} e^{-\nu k^2 \Delta t} \hat{u}_k^{(0)} + \frac{1}{2} \left[\hat{u}_k^{(1)} + \Delta t \left(-\frac{ik}{2} \widehat{\mathrm{tmp}_2}_k + \hat{f}_k^{(1)} \right) \right]$

stable RK3: Shu and Osher

Shu and Osher in general:

$$
\begin{aligned}
u^{(1)} &= u^{(0)} + \Delta t L[u^{(0)}, t^{(0)}] \\
u^{(2)} &= \frac{3}{4}u^{(0)} + \frac{1}{4}u^{(1)} + \frac{1}{4}\Delta t L[u^{(1)}, t^{(1)}] \\
u^{(3)} &= \frac{1}{3}u^{(0)} + \frac{2}{3}u^{(2)} + \frac{2}{3}\Delta t L[u^{(2)}, t^{(2)}]
\end{aligned}
$$

Now it is important to know at what times the $u^{(i)}$ live:

$$
\begin{aligned}
u^{(0)} &: t = 0 \\
u^{(1)} &: t = \Delta t \\
u^{(2)} &: t = \frac{1}{2}\Delta t \\
u^{(3)} &: t = \Delta t
\end{aligned}
$$

Now for our problem:

$$\hat{u}_k^{(1)} = e^{-\nu k^2 \Delta t}\left[\hat{u}_k^{(0)} + \Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(0)^2}\right)}_k + \hat{f}_k^{(0)}\right)\right]$$

$$\hat{\hat{u}}_k^{(2)} = \frac{3}{4}\hat{u}_k^{(0)} + \frac{1}{4}\hat{\hat{u}}_k^{(1)} + \frac{1}{4}e^{\nu k^2 \Delta t}\Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(1)^2}\right)}_k + \hat{f}_k^{(1)}\right)$$

or

$$\hat{u}_k^{(2)} = \frac{3}{4}e^{-\nu k^2 \frac{\Delta t}{2}}\hat{u}_k^{(0)} + \frac{1}{4}e^{\nu k^2 \frac{\Delta t}{2}}\left[\hat{u}_k^{(1)} + \Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(1)^2}\right)}_k + \hat{f}_k^{(1)}\right)\right]$$

$$\hat{\hat{u}}_k^{(3)} = \frac{1}{3}\hat{u}_k^{(0)} + \frac{2}{3}\hat{\hat{u}}_k^{(2)} + \frac{2}{3}e^{\nu k^2 \frac{\Delta t}{2}}\Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(2)^2}\right)}_k + \hat{f}_k^{(2)}\right)$$

or

$$\hat{u}_k^{(3)} = \frac{1}{3}e^{-\nu k^2 \Delta t}\hat{u}_k^{(0)} + \frac{2}{3}e^{-\nu k^2 \frac{\Delta t}{2}}\left[\hat{u}_k^{(2)} + \Delta t\left(-\frac{ik}{2}\widehat{\left(u^{(2)^2}\right)}_k + \hat{f}_k^{(2)}\right)\right]$$

2D Navier-Stokes

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p = \nu \Delta \mathbf{u} + \mathbf{f} \quad , \quad \nabla \cdot \mathbf{u} = 0$$

$\mathbf{u}$ depends only on x, y

$$\Longrightarrow \nabla \times \mathbf{u} = (\partial_x u_y - \partial_y u_x)\mathbf{e}_z = \omega \mathbf{e}_z$$

time evolution for ω:

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \nu \Delta \omega + (\nabla \times \mathbf{f})_z$$

For simplicity forget forcing. Now in Fourierspace:

$$\partial_t \hat{\omega}_{\mathbf{k}} + \widehat{(\mathbf{u} \cdot \nabla \omega)}_{\mathbf{k}} = -k^2 \hat{\omega}_{\mathbf{k}}$$

In Fourierspace we have:

$$\hat{\omega}(\mathbf{k}) = i k_x \hat{u}_y(\mathbf{k}) - i k_y \hat{u}_x(\mathbf{k}) \quad , \quad k_x \hat{u}_x(\mathbf{k}) + k_y \hat{u}_y(\mathbf{k}) = 0$$

And for the velocity:

$$\hat{u}_x(\mathbf{k}) = i k_y \frac{\hat{\omega}(\mathbf{k})}{k^2} \quad , \quad \hat{u}_y(\mathbf{k}) = -i k_x \frac{\hat{\omega}(\mathbf{k})}{k^2}$$

We can write (because of $\nabla \cdot \mathbf{u} = 0$):

$$\mathbf{u} \cdot \nabla \omega = \nabla \cdot (\mathbf{u}\omega)$$

For the pseudospectral method we have to inverse Fouriertransform:

$$\hat{u}_x(\mathbf{k}) \; , \quad \hat{u}_y(\mathbf{k}) \; , \quad \hat{\omega}(\mathbf{k})$$

The rest works as in Burgers !!!